

9. [10 marks]

(2016:ATAR:1)

Two teachers measured the height of the children in their classes. The mean height of children in Class A was 132.8 cm and the mean height of children in Class B was 134.6 cm.

- (a) (i) Name **one** measure that could be used to show the dispersion (spread) of heights in each class. [1]

- (ii) The distribution of height followed a normal curve. State **two** features of a normal curve. [2]

One:

Two:

CONTINUED NEXT PAGE

A statistical test on the difference between the two means showed $p > .05$.

(iii) Outline the conclusion that should be drawn from this result.

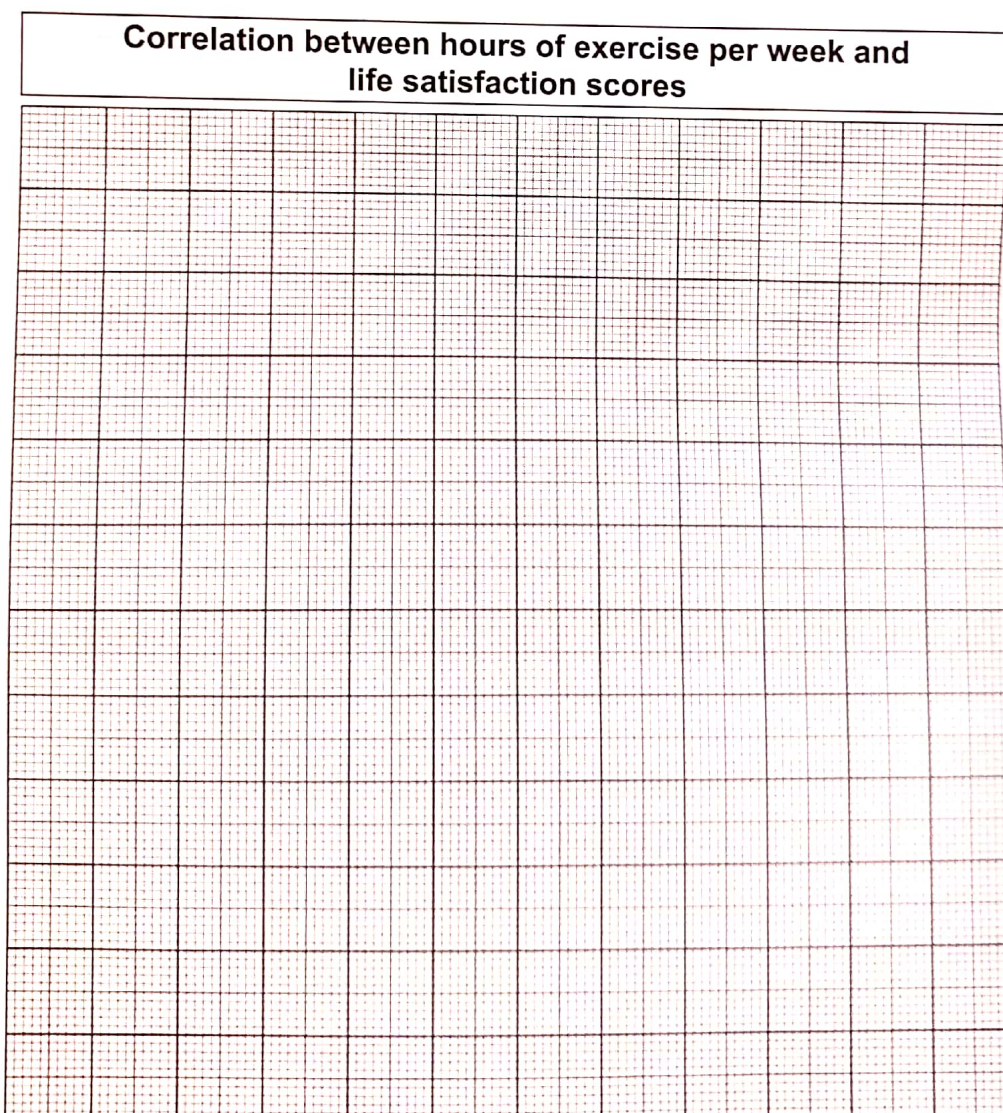
[1]

Researchers were interested in the relationship between exercise and life satisfaction. Data from six participants are shown in the table below.

Participant	Hours of exercise per week	Life satisfaction score
1	2.0	14
2	2.5	16
3	1.5	12
4	5.0	18
5	3.5	17
6	1.0	11

(b) (i) Produce a scatter plot of the data on the grid below.

[4]



- (ii) Outline what the results indicate about the association between exercise and life satisfaction. [2]

10. [8 marks]

(2016:ATAR:2)

Researchers wanted to investigate whether listening to music while studying affects memory in adolescents. All Year 12 students at Grand High School agreed to participate in the research. The boys and girls were divided into separate groups. Both groups were given 10 minutes to study a list of words. The girls listened to music whilst studying and the boys did not. The girls were then given a page of words and had to identify those that they had studied. The boys were given five minutes to write down as many of the words as they could remember.

- (a) Identify whether the participants were a sample or a population and provide **one** reason for your response. [2]

- (b) Identify whether this research was experimental or non-experimental and provide **one** reason for your response. [2]

- (c) Complete the table to identify **two** sources of error in the data and state one way of reducing each. [4]

Source of error	Way of reducing error

11. [12 marks]

(2016:ATAR:3)

- (a) (i) State **one** way in which researchers can maintain participants' rights to anonymity. [1]

- (ii) State **two** ways in which researchers can maintain participants' rights to confidentiality. [2]

One: _____

Two: _____

Researchers wanted to study children's emotional development. They recruited a sample of two-year-old children and planned to test them every two years until the age of 12.

- (b) Name the type of study design referred to in this statement. [1]

- (c) Name and describe an alternative study design that the researchers could use. [3]

- (d) Outline **one** advantage of the study design named in part (b). [2]

Researchers wanted to study how adolescents respond to stressful situations. Participants were told that they would be required to give a two-minute speech to an audience and then their levels of stress would be measured.

- (e) (i) Name **one** example of an objective quantitative measure that the researchers could use. [1]

- (ii) Name **one** example of a subjective quantitative measure that the researchers could use. [1]

- (iii) State **one** reason why an objective quantitative measure might produce data that are more valid than a subjective quantitative measure in this study. [1]

12. [17 marks]

(2017:ATAR:1)

Two groups of psychology students planned to study optimism in young people. One group planned to use a quantitative measure and the other group a qualitative method of data collection.

- (a) Name the type of subjective quantitative measure shown below.

[1]

	Strongly agree	Agree	Neither agree nor disagree	Disagree	Strongly disagree
I always look for the positive in any situation	☐	☐	☐	☐	☐

- (b) Name and describe **one** example of a qualitative method that could be used in this research.

[3]

- (c) Outline **two** advantages of using a quantitative measure.

[2]

One: _____

Two: _____

- (d) Outline **two** advantages of using a qualitative method.

[2]

One: _____

Two: _____

- (e) Name and outline **one** way in which the researchers could test the reliability of the subjective quantitative measure. [2]

- (f) (i) If the researchers collected information about participants' favourite type of music to investigate whether people who like different music have different levels of optimism, name the type of study this would be. [1]

- (ii) Identify **one** limitation of this type of study. [1]

- (g) (i) If participants' optimism was calculated on a scale from 0 to 20, name **one** measure that would be suitable for showing the central tendency in the scores. [1]

- (ii) If participants' favourite type of music was measured using a checklist with four different categories (e.g., Rock, Jazz, Rap, Classical), name **one** measure that would be suitable for showing the central tendency in the scores. [1]

- (h) A statistical test on the data showed that participants who preferred Rock music had the highest optimism scores ($p < .05$). Outline what $p < .05$ means. [3]

13. [19 marks]

Psychologists designed a study to test whether drinking goji berry juice improved adults' energy levels. Participants were randomly divided into two groups. Participants in Group 1 drank goji berry juice twice a day for 14 days. Participants in Group 2 drank artificially coloured and flavoured water that looked and tasted like goji berry juice twice a day for 14 days. After the 14 days, the participants were asked to rate their energy levels on a scale from 0 to 100.

- (a) (i) Identify the independent variable in the study. [1]

- (ii) Identify the dependent variable in the study. [1]

- (b) (i) State the name that would be given to Group 1 in the study. [1]

- (ii) State the name that would be given to Group 2 in the study. [1]

- (c) Write an operational hypothesis for this study. [3]

- (d) (i) State the psychological term for the flavoured water in this study. [1]

- (ii) Explain why participants in Group 2 were given the flavoured water to drink. [2]

- (e) (i) Identify the main source of error in the data in this study. [1]

- (ii) Outline **one** way of improving the study to reduce this source of error. [1]

- (f) Identify **two** things that the researchers must explain to participants at the start of the study to ensure that the research is conducted ethically. [2]

One: _____

Two: _____

- (g) (i) State **one** thing that the researchers must explain to participants at the end of the study to ensure that the research is conducted ethically. [1]

- (ii) Explain the reason for your response to (g)(i). [2]

- (h) Outline **two** things that the researchers should do to maintain confidentiality of participants' data. [2]

One: _____

Two: _____

14. [11 marks]

Researchers wanted to investigate the ongoing health effects of working as a medical student at a busy city hospital. Participants were first year medical students chosen from a hospital close to the university where the researchers worked. They were followed throughout their medical training. Researchers met participants once a year over an eight-year period.

(a) Name the study design used by researchers.

[1]

(b) State **one** advantage of using this type of study design.

[1]

(c) State **two** disadvantages of using this type of study design.

[2]

One: _____

Two: _____

(d) The researchers state that they have used convenience sampling for this study.

(i) What evidence is given to support this statement?

[1]

(ii) State **one** disadvantage of this type of sampling.

[1]

(e) When researchers met participants, they were interested in collecting quantitative data to measure health effects. Name **one** subjective quantitative measure and **one** objective quantitative measure that might have been used in this study.

[2]

Quantitative collection method	Measure
Subjective	
Objective	

- (f) Identify **three** roles the researchers would have had in conducting this study. [3]

One: _____

Two: _____

Three: _____

15. [16 marks]

(2018:ATAR:2)

Psychologists designed a study to determine the effects of rosemary oil on cognitive ability in high school students. A total of 100 Year 12 students were involved, 40 from School A and 60 from School B.

Before the study began, all 100 participants completed a standardised cognitive ability test and the results were collected and analysed. Students from School A were instructed to take two rosemary oil tablets a day, one in the morning and one at night, for three months. Students from School B were not given the rosemary oil tablets.

After three months, students from each school sat the same standardised cognitive ability test. Standard deviations are shown in the table below.

Statistical tests on the data showed that $p > 0.05$.

Standard deviations of School A and School B

	Prior to the study		End of the study	
	Mean	Standard deviation	Mean	Standard deviation
School A	72	1.5	75	1.5
School B	73	2.5	76	2.5

- (a) In psychological research, what does the standard deviation tell us about data? [1]

- (b) State **one** advantage of using standard deviation in data analysis. [1]

- (c) School A has a lower standard deviation than School B. Describe the variability of the data from each school. [2]

- (d) Researchers wanted to visually display the distribution of data, including the mean and standard deviation from the mean. Name the graph that might have been used to show this. [1]

- (e) In the raw data, School A had a range of 12 and School B had a range of 22. What is meant by the term 'range'? [1]

- (f) (i) Define 'uncontrolled variables' in psychological research. [1]

- (ii) There were uncontrolled variables in this study. State **two** uncontrolled variables that might have affected the results of the study. [2]

One: _____

Two: _____

- (g) (i) Experimental research must be evaluated in order to determine whether the results can be generalised to the wider population. State **three** conditions that must be met in order to generalise findings and draw conclusions. [3]

One: _____

Two: _____

Three: _____

- (ii) Can the findings from this research study be generalised to the wider population? Provide **three** reasons for your answer. [4]

Circle either Yes or No

One: _____

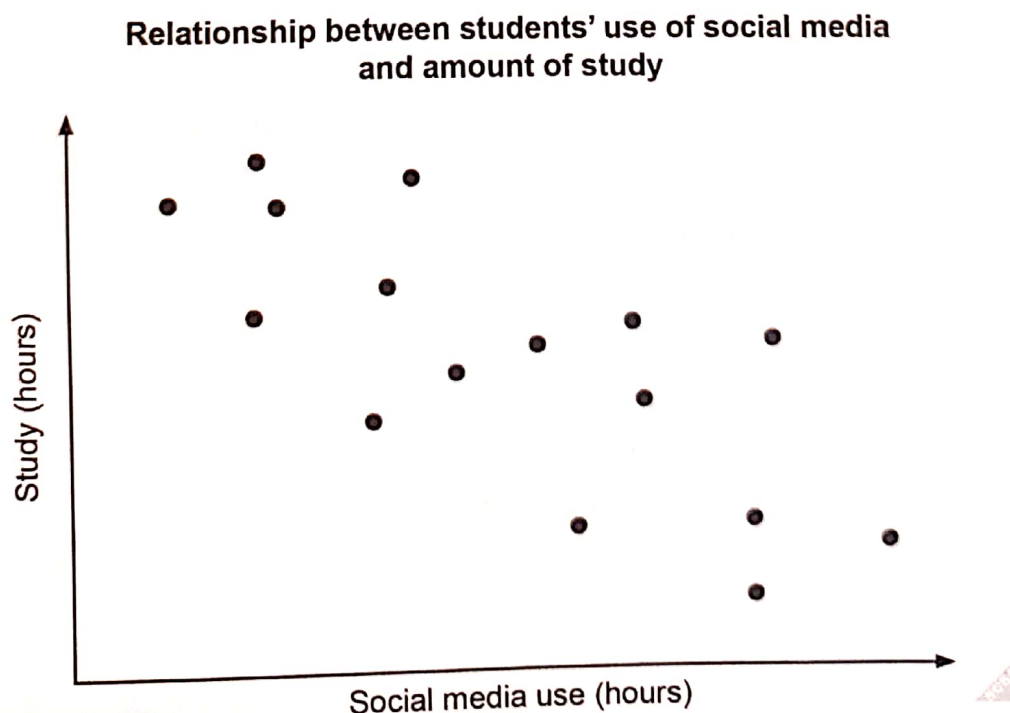
Two: _____

Three: _____

16. [6 marks]

(2018:ATAR:3)

A psychology professor researched the relationship between students' use of social media and the amount of study they complete. The results are shown in the scatter plot below.



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(a) Identify the type of non-experimental research method used.

[1]

(b) State the strength and direction of the relationship in the scatter plot above.

[2]

(c) Describe how this research design differs from an experimental research design.

[2]

(d) The professor wanted to ensure that the findings from this research were valid. Define the term 'validity' as used in psychological research.

[1]

17. [13 marks]

(2019:ATAR:1)

At a university, 40 first year psychology students participated in a study examining the effects of loud music on typing accuracy. Students were given a typing test to complete, with 20 students listening to loud music through headphones and 20 students without music.

At the end of the test, the total number of mistakes was calculated for each condition.

(a) (i) Identify the independent variable in this study.

[1]

(ii) Identify the dependent variable in this study.

[1]

- (iii) Identify **two** uncontrolled variables in relation to the participants in this study. [2]

One: _____

Two: _____

- (b) Write an operational hypothesis for this study. [4]

The psychology students were asked to write a report that outlined information about the sample and population.

- (c) How do sample and population differ? [2]

After completing the study, participants were asked to fill out a short, fixed-response (ranging from strongly agree to strongly disagree) questionnaire about how important they thought music was to different aspects of their lives.

- (d) (i) Name this type of subjective quantitative measure. [1]

- (ii) State **one** advantage and one disadvantage of using this type of subjective quantitative measure. [2]

Advantage: _____

Disadvantage: _____

18. [12 marks]

(2019:ATAR:2)

Nena conducted a series of semi-structured interviews with 20 Year 12 students to find out their opinions about changing the legal drinking age.

(a) (i) What kind of method of data collection has Nena used? [1]

Circle: Qualitative OR Quantitative

(ii) Give **one** reason for your response. [1]

(b) Compare qualitative and quantitative methods of data collection in the table below. [4]

Method of data collection	Type of data collected	Advantage of data collection method
Qualitative		
Quantitative		

(c) State **one** alternative method of data collection Nena could have used for this study. [1]

In addition to conducting her interviews, Nena had the students complete a survey asking them what age they thought drinking should be legalised. The raw data for the 20 participants was:

17, 17, 16, 18, 19, 20, 16, 15, 22, 21, 20, 19, 18, 17, 18, 18, 20, 21, 18, 19.

- (d) Create a frequency table based on this data in the space below. [5]